Geocomputation

Spatial Models



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Linear Models
- Spatial Models
- Assessment: Social Atlas

Before we start

- Go to www.menti.com
- Use code: XXXX XXXX

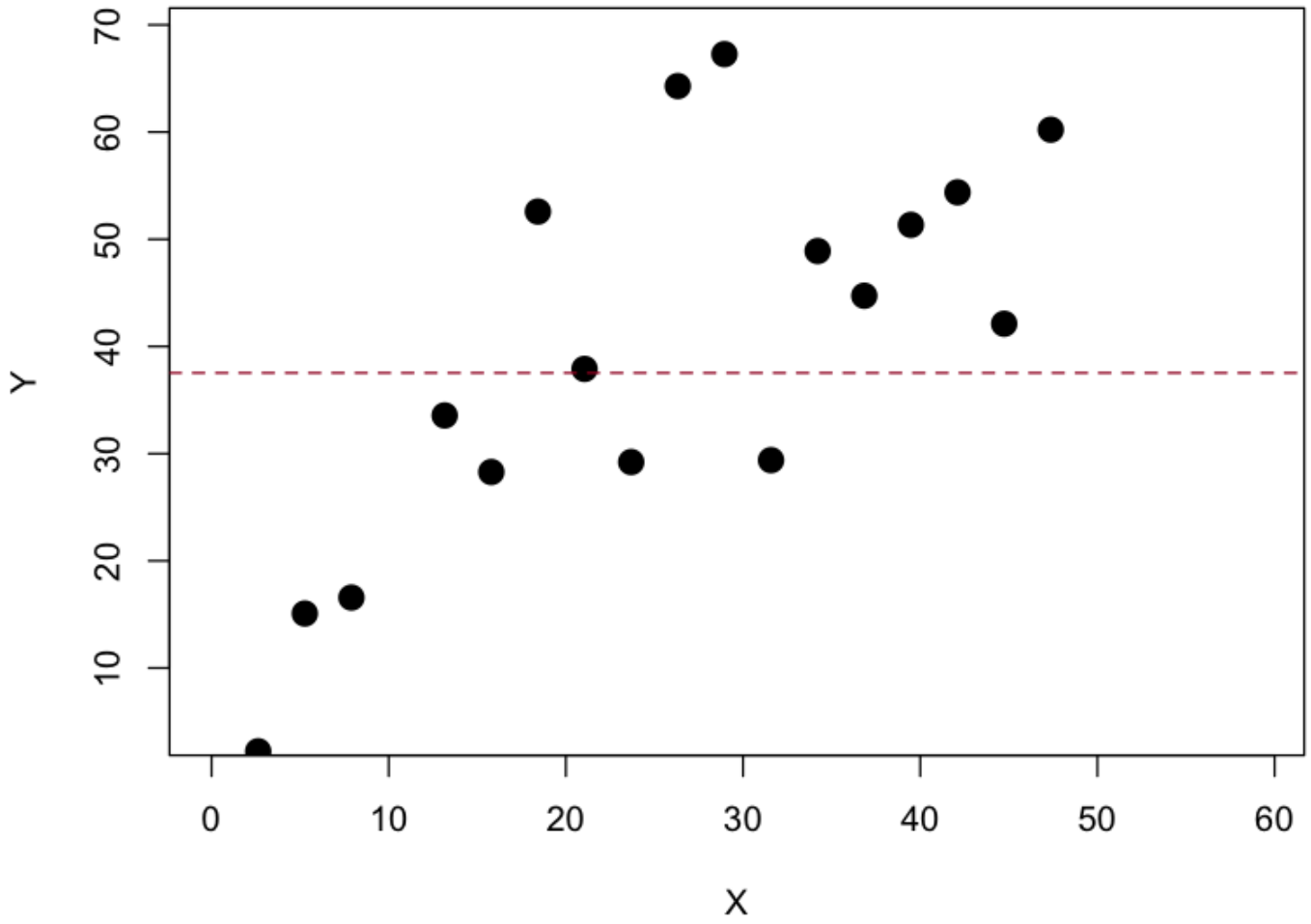
Bivariate regression

- We often want to know more than just whether two variables are related; we also want to describe and predict how one variable changes as another variable changes.
- To do this we can use a regression model to examine the relationship between a dependent variable (y) and one or more independent variables (x).

Bivariate regression

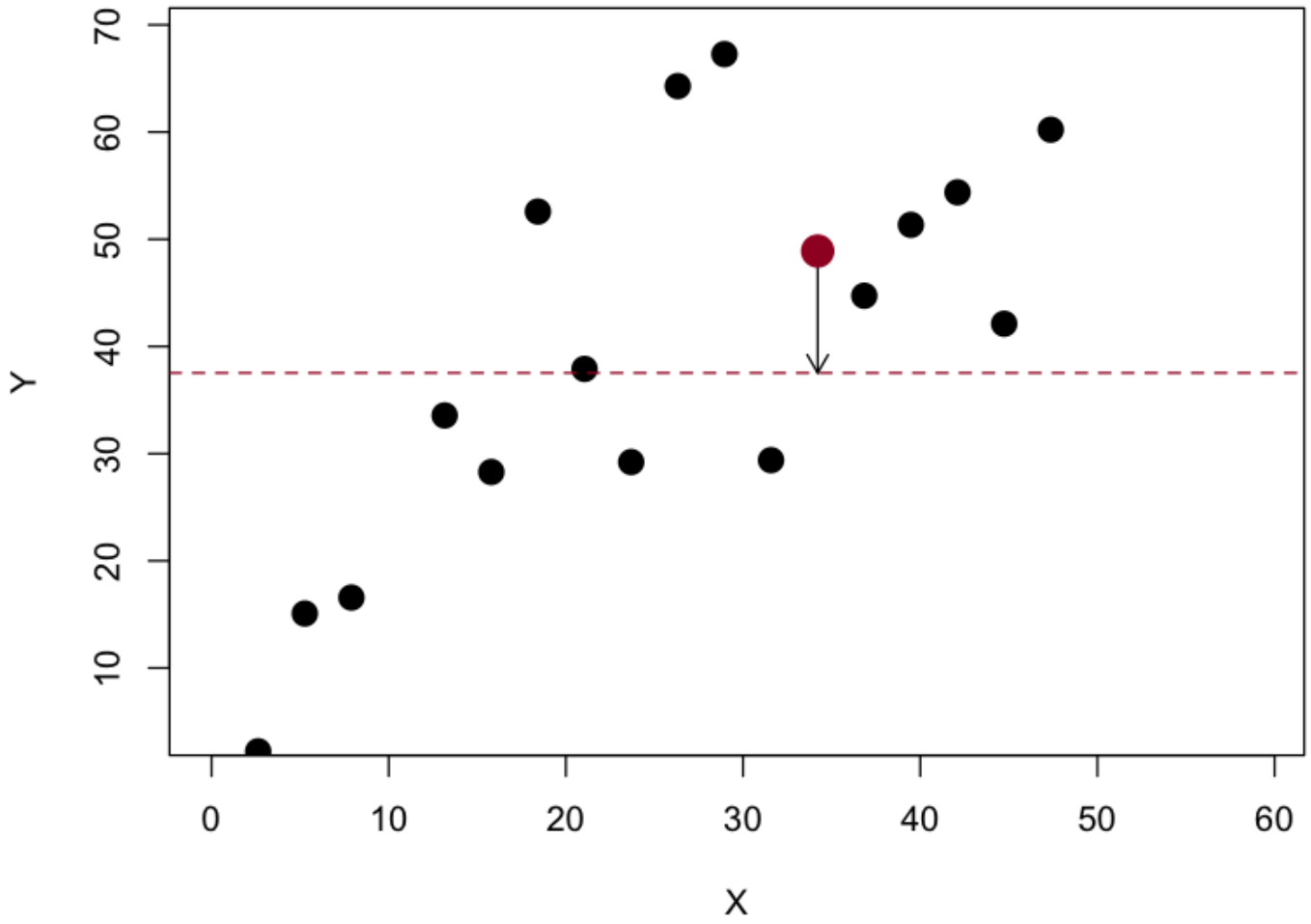
- Linear regression uses a line to summarise the relationship between x and y.
- The aim is to find the line which best represents the relationships in the data.
- Typically, this line will not pass through every data point meaning we cannot predict y exactly.

Null model



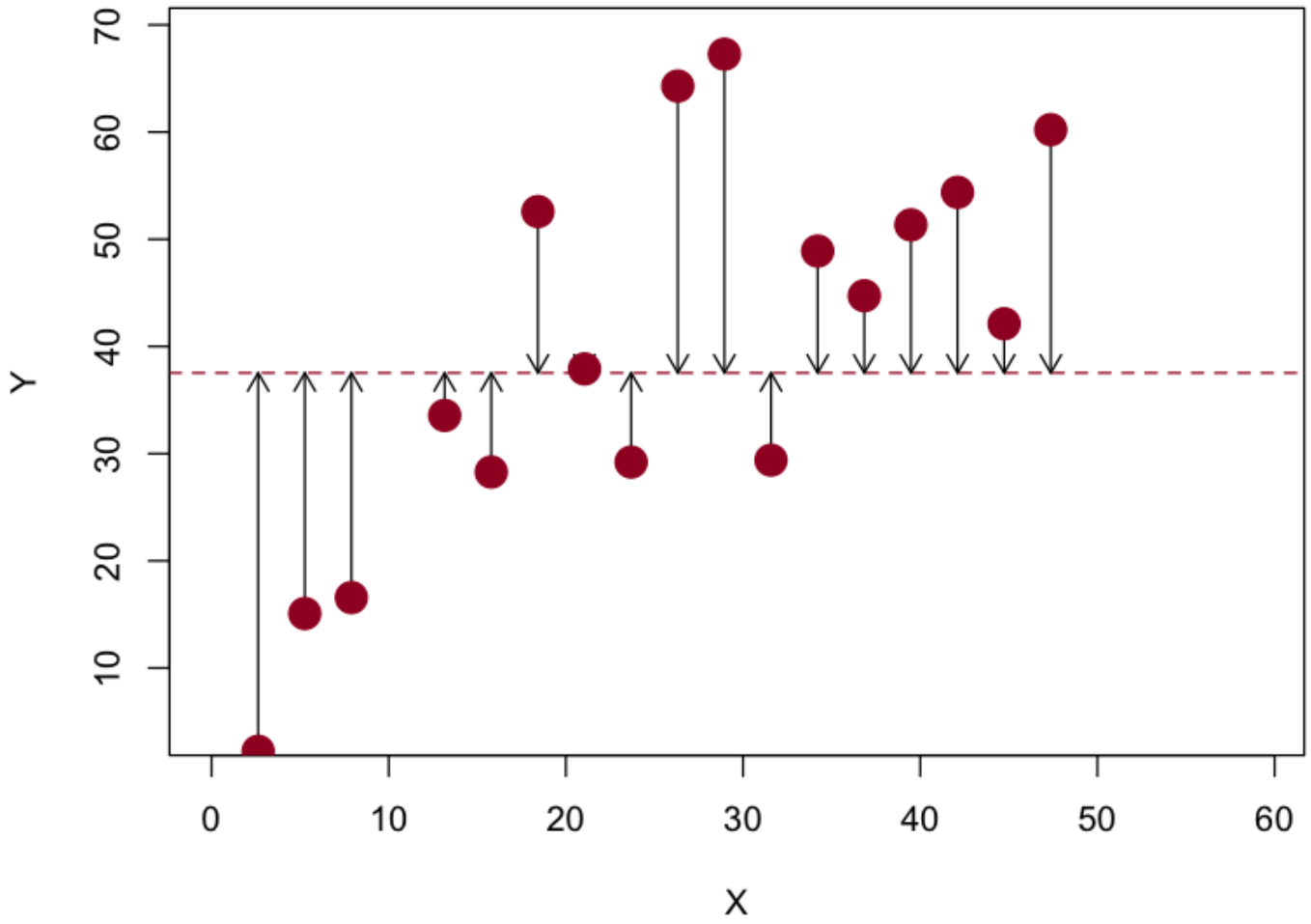
With no information about x, the best prediction of y is its mean.

Sum of squared errors



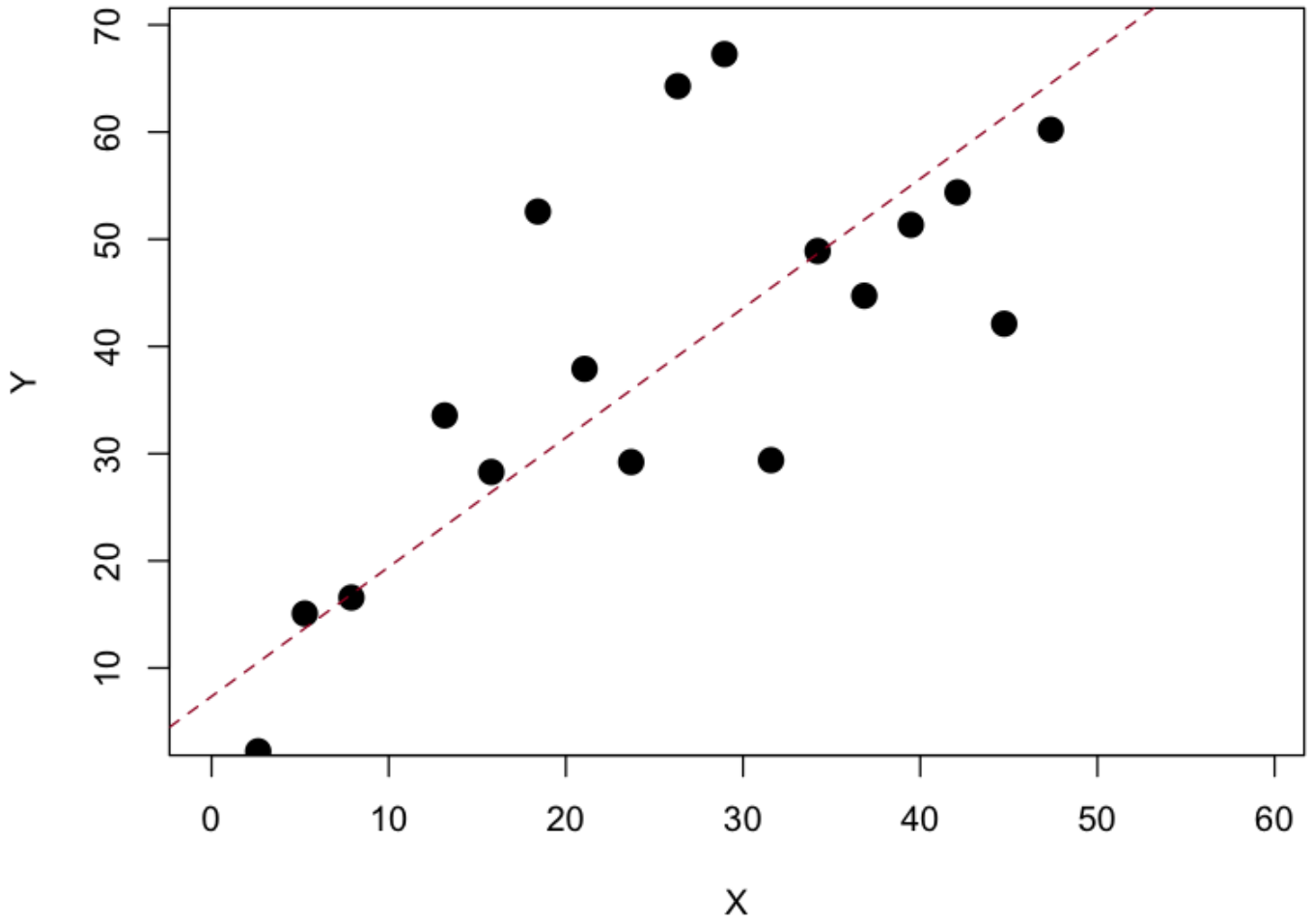
Error: the gap between an observed value and its prediction.

Sum of squared errors



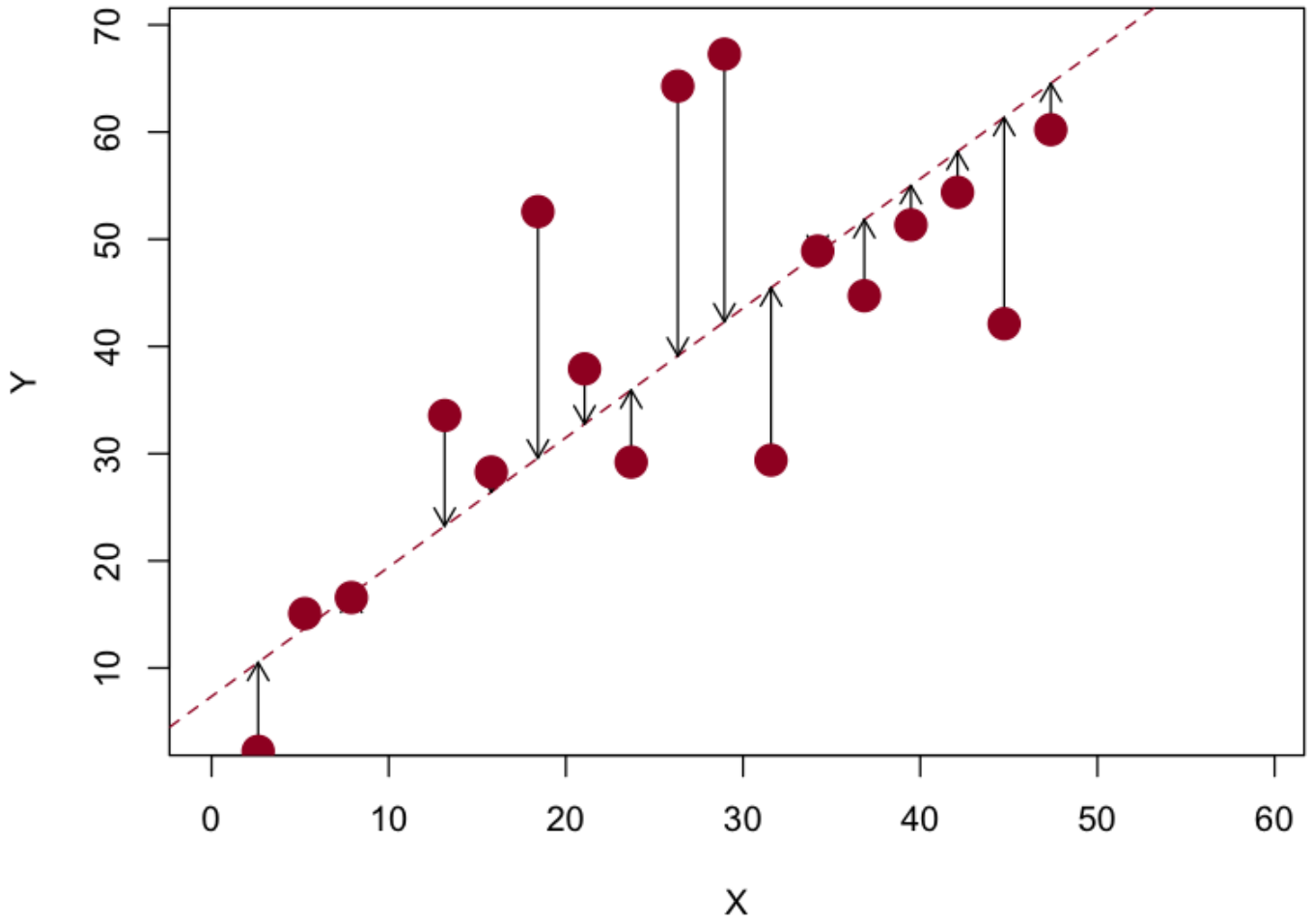
Square the errors and add them up: the sum of squared errors (SSE).

Minimise sum of squared errors



A line that follows the points leaves smaller errors.

Minimise sum of squared errors



Ordinary least squares picks the line with the smallest SSE.

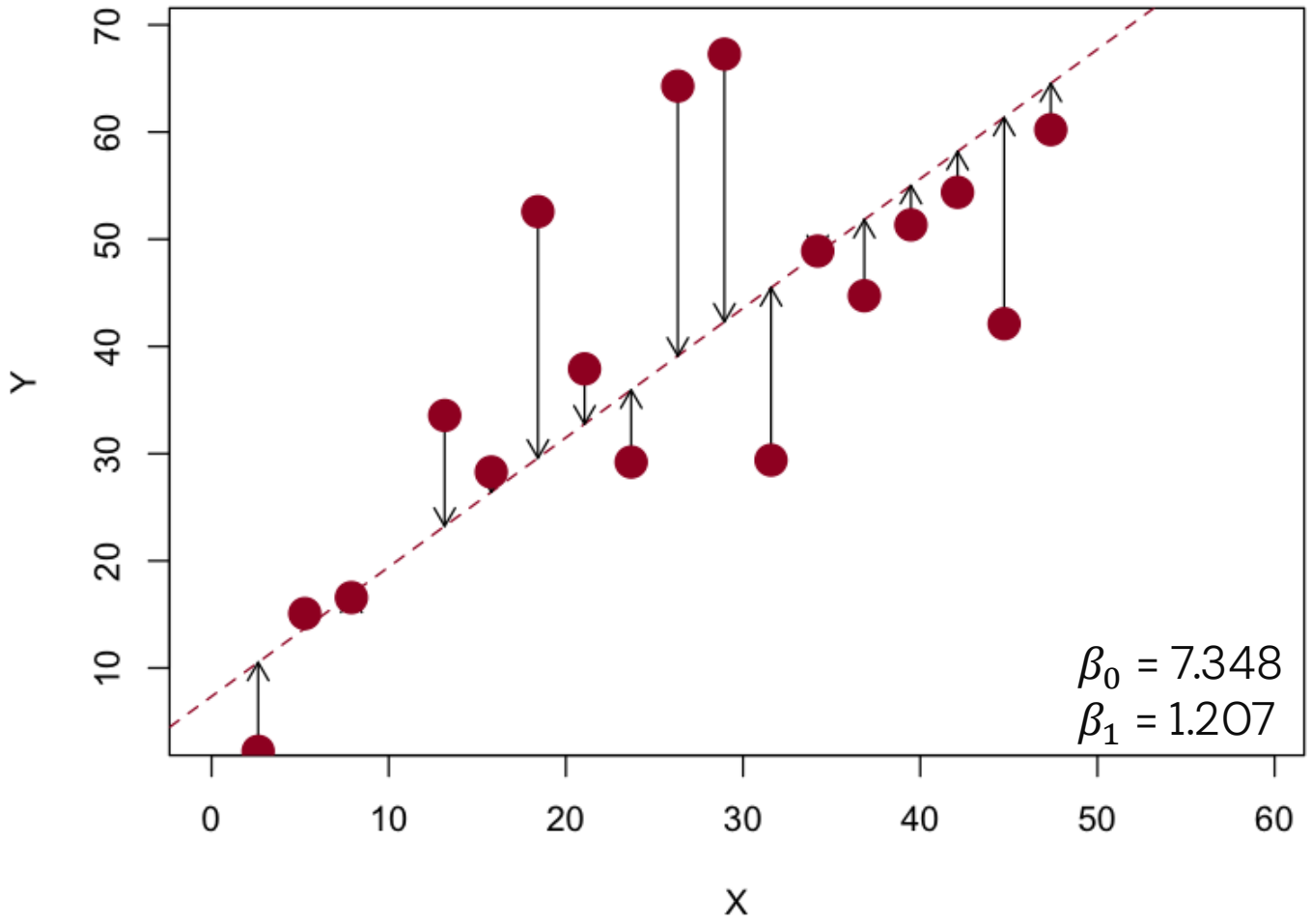
Bivariate regression

- Ordinary Least Squares (OLS) regression:

$$\hat{y} = \beta_0 + \beta_1 x$$

- The β terms are coefficients that define the regression line.
- The model estimates these parameters to find the line that gives the smallest sum of squared errors: Ordinary Least Squares (OLS) regression.

Minimise sum of squared errors



Each one-unit increase in x goes with a 1.207 increase in y, on average.

Model fit

- The R^2 measures the proportion of the variance in the dependent variable that is explained by the model.
- Values range from 0 to 1.
  - 0: the model explains none of the variance.
  - 1: the model explains all the variance.
- R^2 compares the errors of our line with the errors of the mean-only model.

Multiple regression

- Multiple regression uses several predictors at once, because it is often unrealistic to attribute variation in y to a single x.
- Each coefficient shows the association between that predictor and y while the other predictors are held constant.

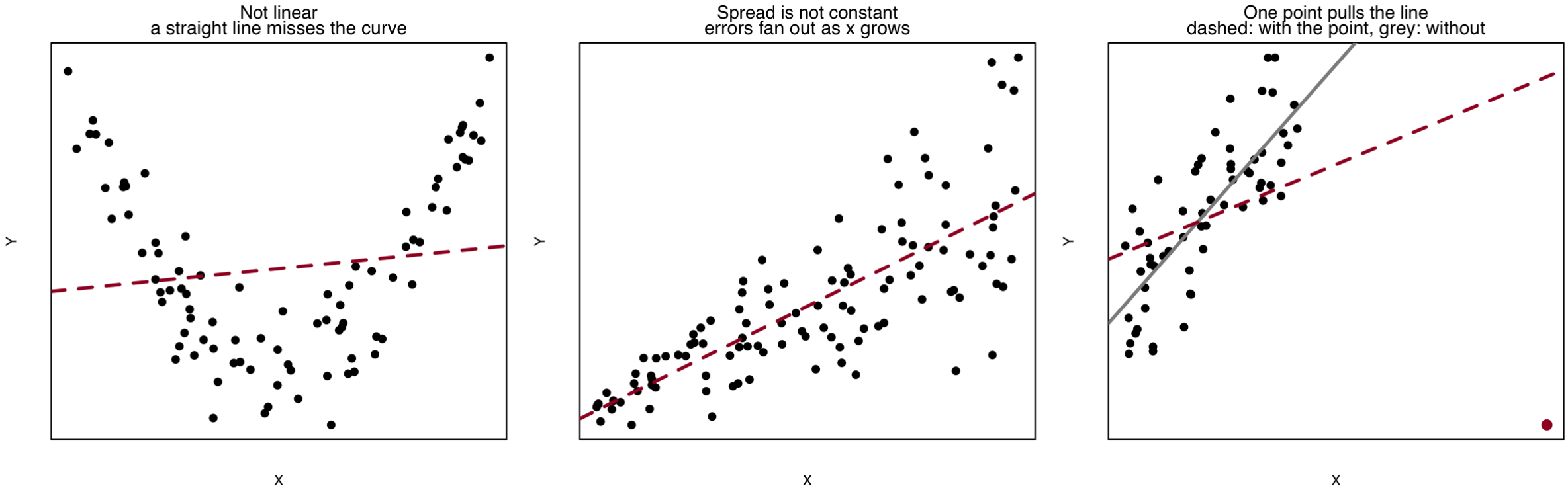
Which regression for which outcome

Distribution of dependent variable	Example	Suitable model
Continuous	Income	Linear regression
Binary	Employment status	Logistic regression
Binomial	Proportion of homeowners	Logistic regression
Count	Number of crimes	Poisson regression

Assumptions

- Linearity: each extra unit of x adds the same amount to y.
- Independence: the errors are unrelated.
- Homoscedasticity: the errors have a similar spread everywhere.
- Normality of errors: the errors are roughly bell-shaped.
- No multicollinearity: the predictors are not strongly correlated with each other.
- No influential observations: no single place drives the result.

Assumptions: what a problem looks like

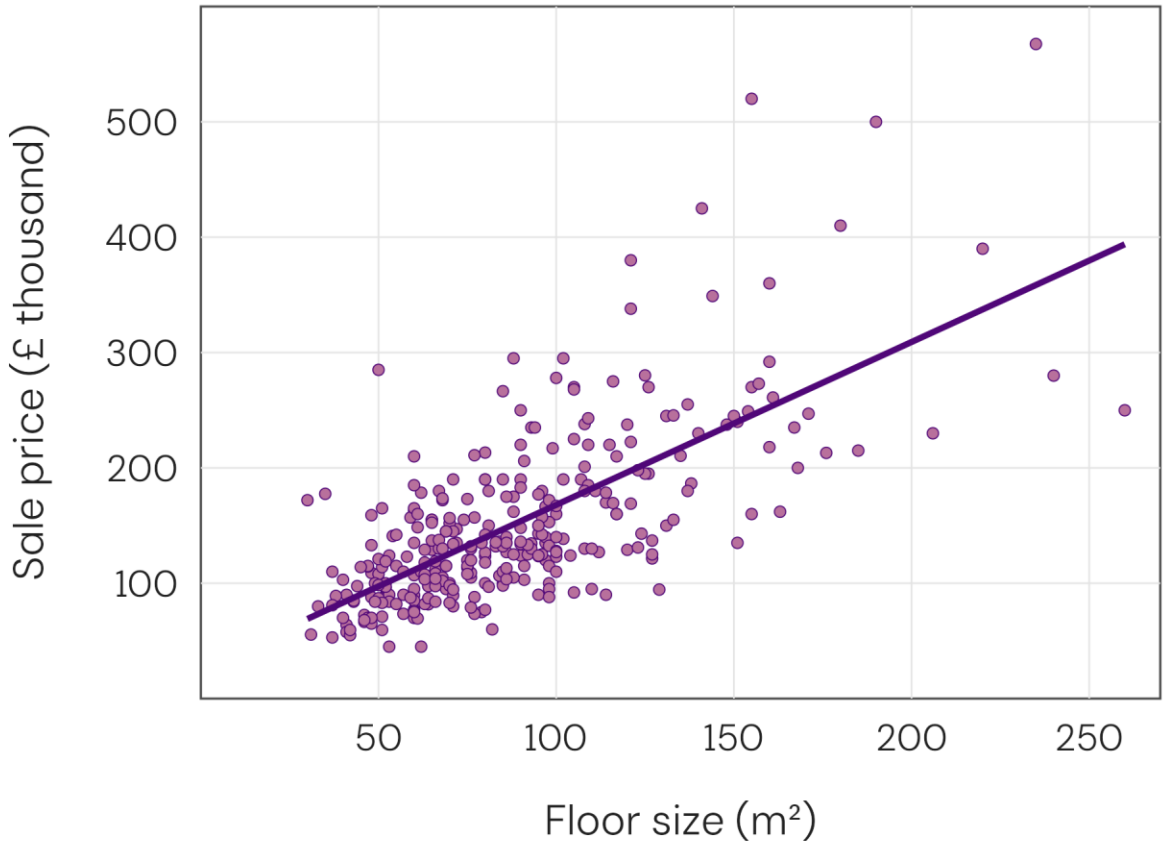


Plot the data and the errors: patterns like these tell us when to worry.

Multicollinearity

- Multicollinearity means that predictors are strongly correlated with each other.
- This makes the coefficients unstable and hard to interpret.
- The variance inflation factor (VIF) measures it: below 5 is fine, 5 to 10 is moderate, above 10 is a problem.

One slope for the whole of London



One line for all of London

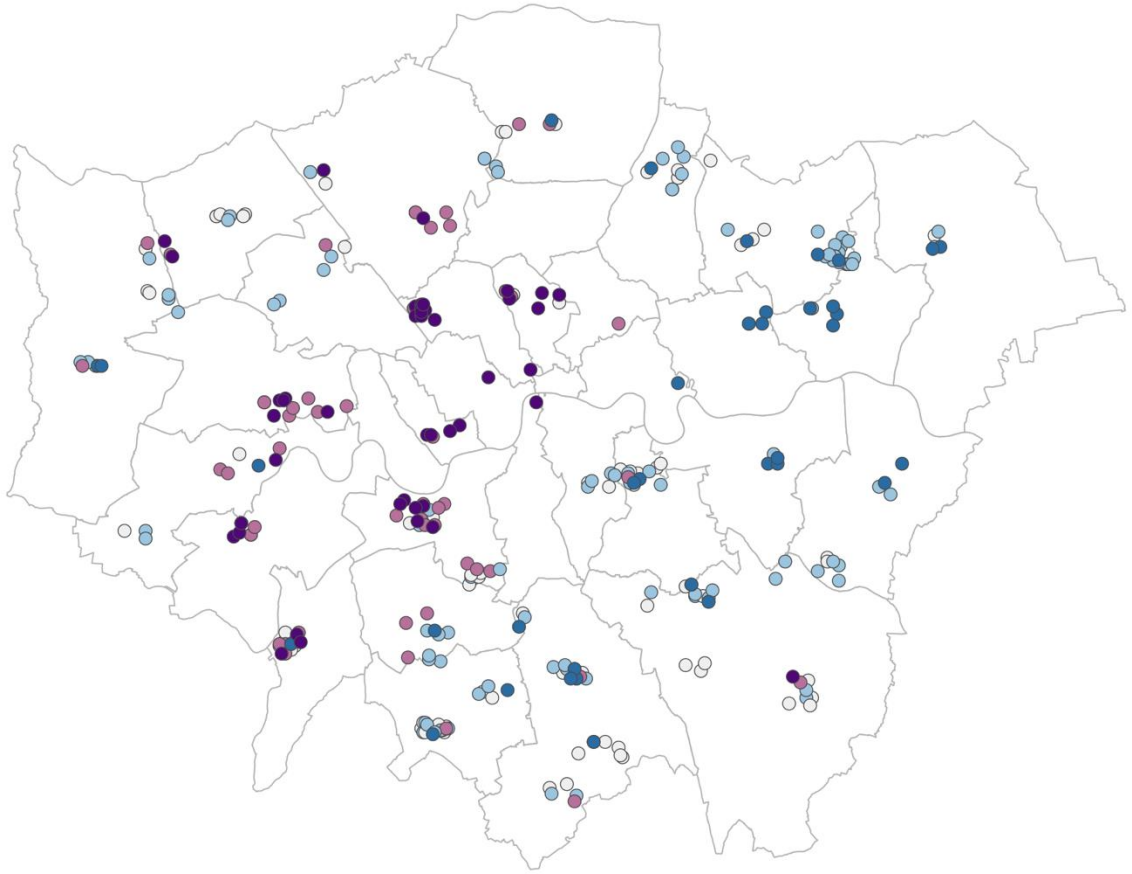
Each extra square metre goes with about £1,412 more.

In the centre and in the outer boroughs alike.

$R^2 = 0.49$

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

Residuals can cluster in space



OLS residual (£ thousand)

- below −50
- −50 to −15
- −15 to 15
- 15 to 50
- above 50

Purple: sold for more than the line predicts.

Blue: sold for less.

Moran's I of the residuals:
0.45 ($p < 0.01$)

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

Checking the residuals

When building a model based on spatial data:

- Map the residuals of the linear model to visually inspect for spatial patterns.
- Calculate Moran's I statistic on the residuals to assess spatial autocorrelation.
- If spatial autocorrelation is present, fit a spatial model to account for it.
- Recalculate Moran's I statistic on the residuals of the spatial model to confirm that the autocorrelation has been addressed.

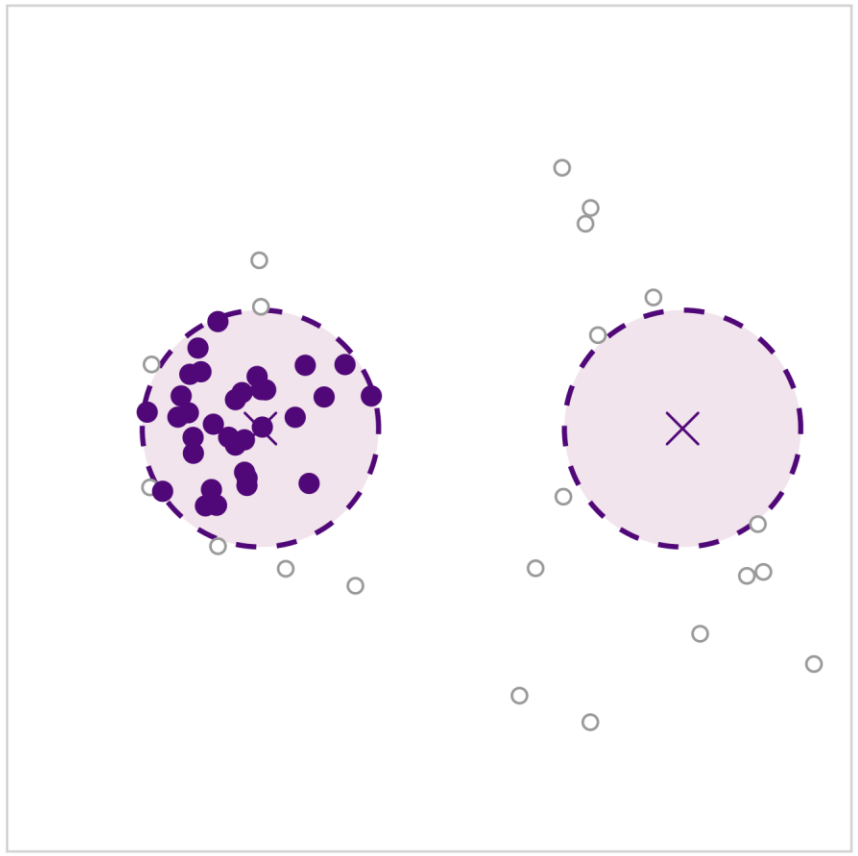
Other spatial models

- The spatial error model allows the errors of neighbouring areas to be related, for example because a spatial variable is missing.
- The spatial lag model lets the outcome in one area depend on the outcomes of its neighbours.
- Both keep one set of coefficients for the whole study area.
- In R, the `spatialreg` package fits both.

Geographically weighted statistics

- Unlike traditional global models, which estimate a single set of parameters for the entire study area, geographically weighted statistics allow for parameter estimates that vary across different locations.
- Local means, standard deviations, correlations and regression coefficients.
- Typically uses some kernel function to weigh observations based on their distance from the location of interest.

Fixed kernel: the same distance everywhere



× regression point

● point inside the window

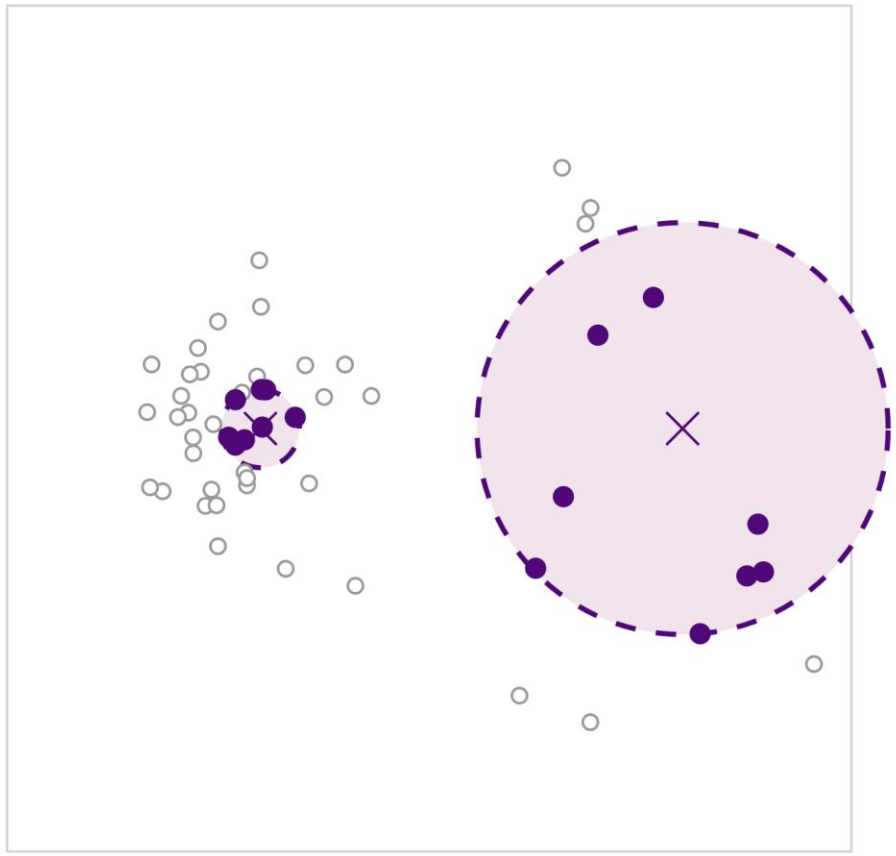
○ point outside

Same radius everywhere.

Dense area: 33 points in the window.

Sparse area: 0 points in the window.

Adaptive kernel: the same number of neighbours



× regression point

● point inside the window

○ point outside

Same number of points: 8.

Dense area: a small window.

Sparse area: a window 5.3 times wider.

Bandwidth: how local is local?

- The bandwidth sets how many observations count as local.
- Small bandwidth: very local estimates, but noisy.
- Large bandwidth: smooth estimates that come close to one global model.
- A common choice: the bandwidth with the best cross-validation score.

Geographically weighted regression

- These ideas can be extended to correlation and regression:
  - Geographically weighted correlation (GWC)
  - Geographically weighted regression (GWR)
- The basic GWR equation is:

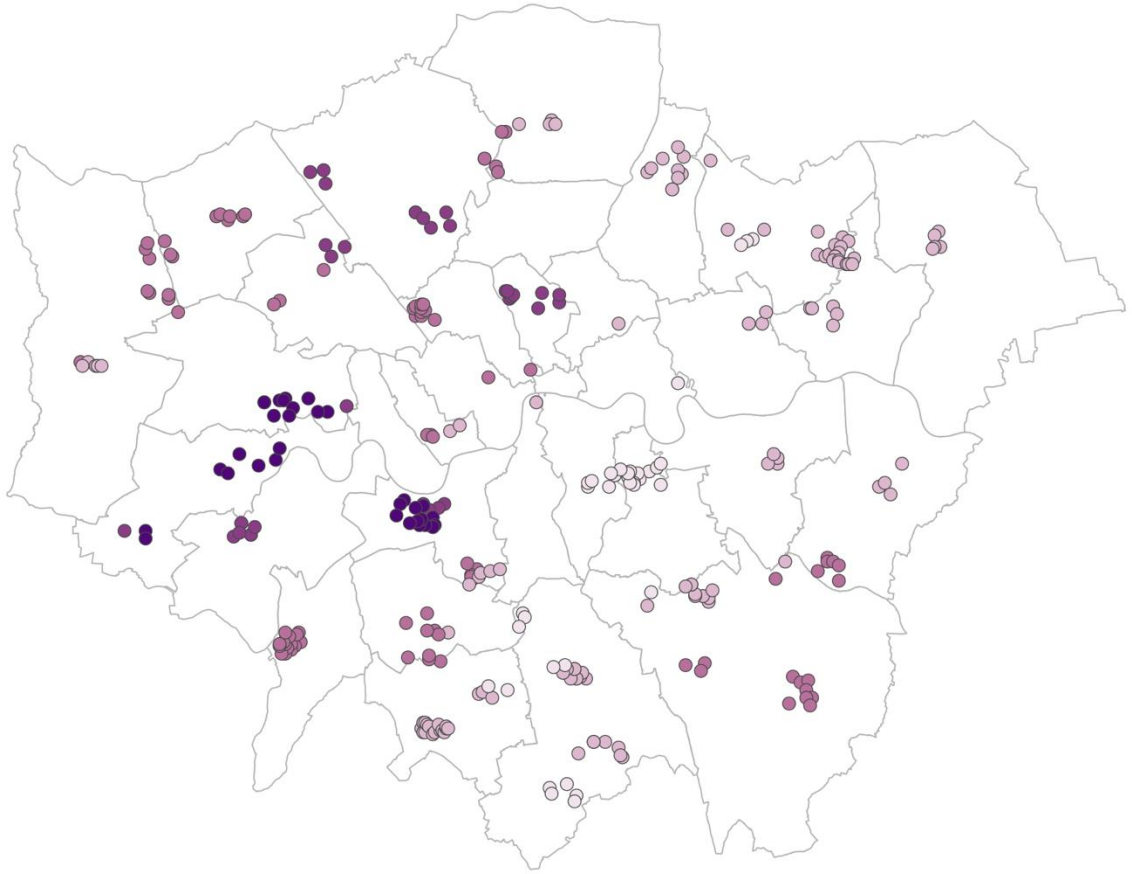
$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^{p} \beta_k(u_i, v_i) x_{ik} + \epsilon_i$$

where (u_i, v_i) are the coordinates of location i and $\beta k(u_i, v_i)$ are the location-specific coefficients.

Geographically weighted regression

- Each area has its own set of regression coefficients.
- Each location has its own R^2 value.
- Each area has its own standard error for each coefficient.
- More recently: bandwidths can vary between different variables.

Local slopes: the price of space varies by place



Local slope (£ per m²)

- below 1,000
- 1,000 to 1,400
- 1,400 to 2,000
- 2,000 to 2,600
- above 2,600

Local slopes run from £762 to £3,079.

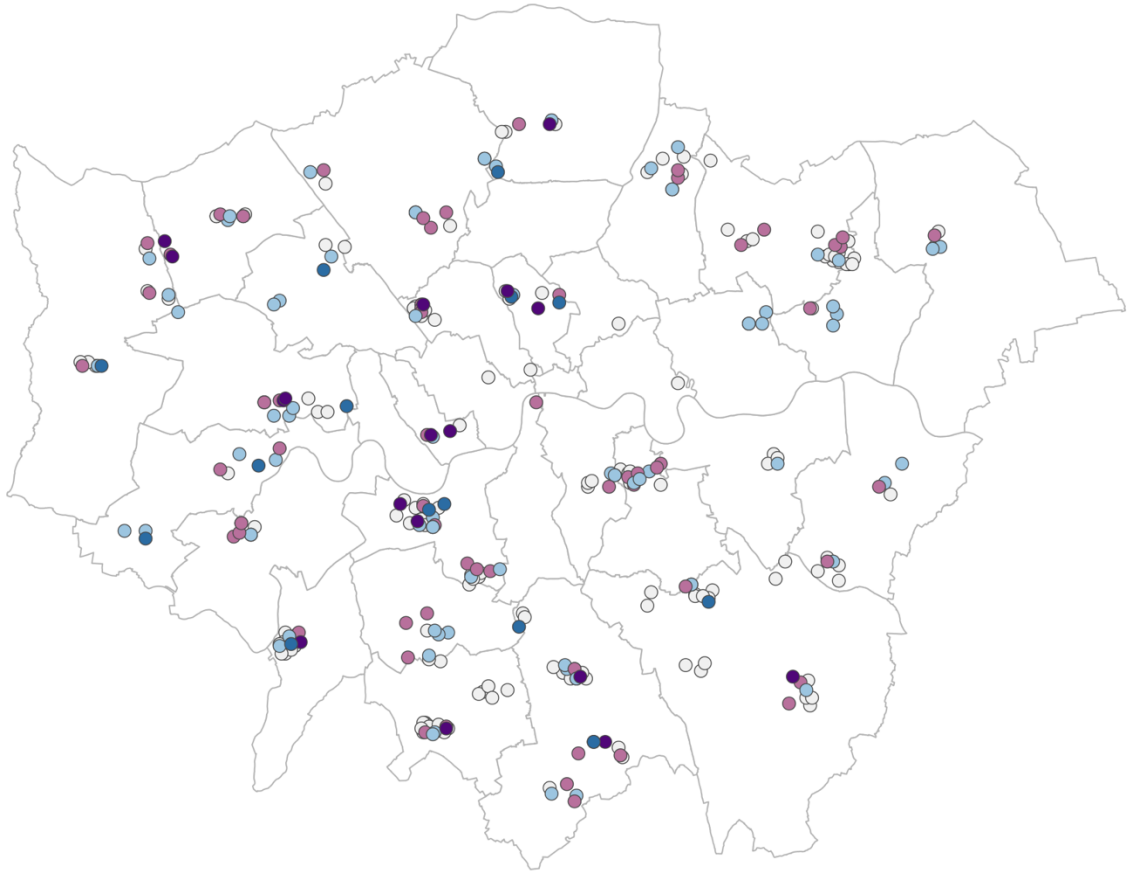
The global slope was £1,412.

Local R^2:

0.54 to 0.95

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

After GWR: the residuals no longer cluster



GWR residual (£ thousand)

- below −50
- −50 to −15
- −15 to 15
- 15 to 50
- above 50

R^2 rises from 0.49 to 0.84.

No clusters of one colour remain.

Moran's I of the residuals:

−0.03 (p = 0.81)

Source: GWmodel example data (LondonHP): 316 house sales in London. Prices in pounds.

Reading a coefficient map

- A coefficient map shows where the association is stronger or weaker.
- It does not show where the association is significant: check the local t-values.
- With hundreds of overlapping local tests, treat significance as a guide.
- Strongly correlated predictors can make local coefficients swing in sign: check multicollinearity first.

Conclusion

- Spatial autocorrelation in the residuals breaks the independence assumption: the standard errors and p-values become unreliable.
- Spatial error and spatial lag models keep one set of coefficients for the whole study area.
- GWR lets the coefficients vary by place and can reduce the clustering in the residuals.
- Always map the residuals and test them again with Moran's I, before and after.

Questions

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Assessment

- Social Atlas Coursework Assessment (60%): The first assessment will involve the completion of a spatial analysis project, based on the theory, concepts and application learnt during the module. For this coursework you are required to create a small *social atlas* on a topic or area that interests you. Deadline: TBC.
- On Moodle: guidance as well as examples from previous years.

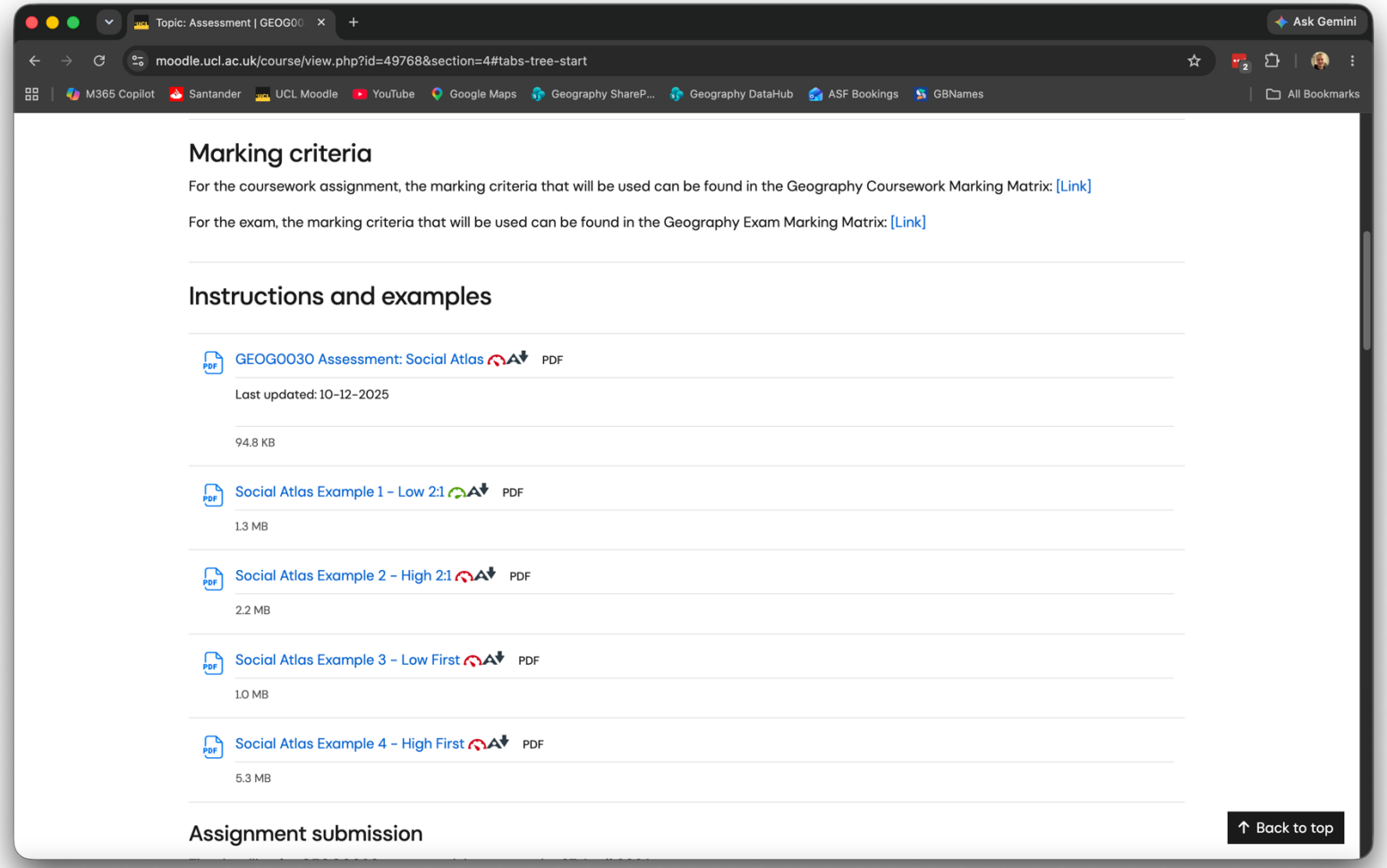
Assessment

- You should create a minimum of 4 maps and a maximum of 6.
- You should create a minimum of 2 graphs / charts and a maximum of 4.
- You can choose a specific theme or zoom into a particular area.
- You should aim to utilise a range of different techniques taught in the Geocomputation module to explore your topic – but make sure you apply the techniques in appropriate manner and with the right data types.
- Greater London cannot be used as a case study.

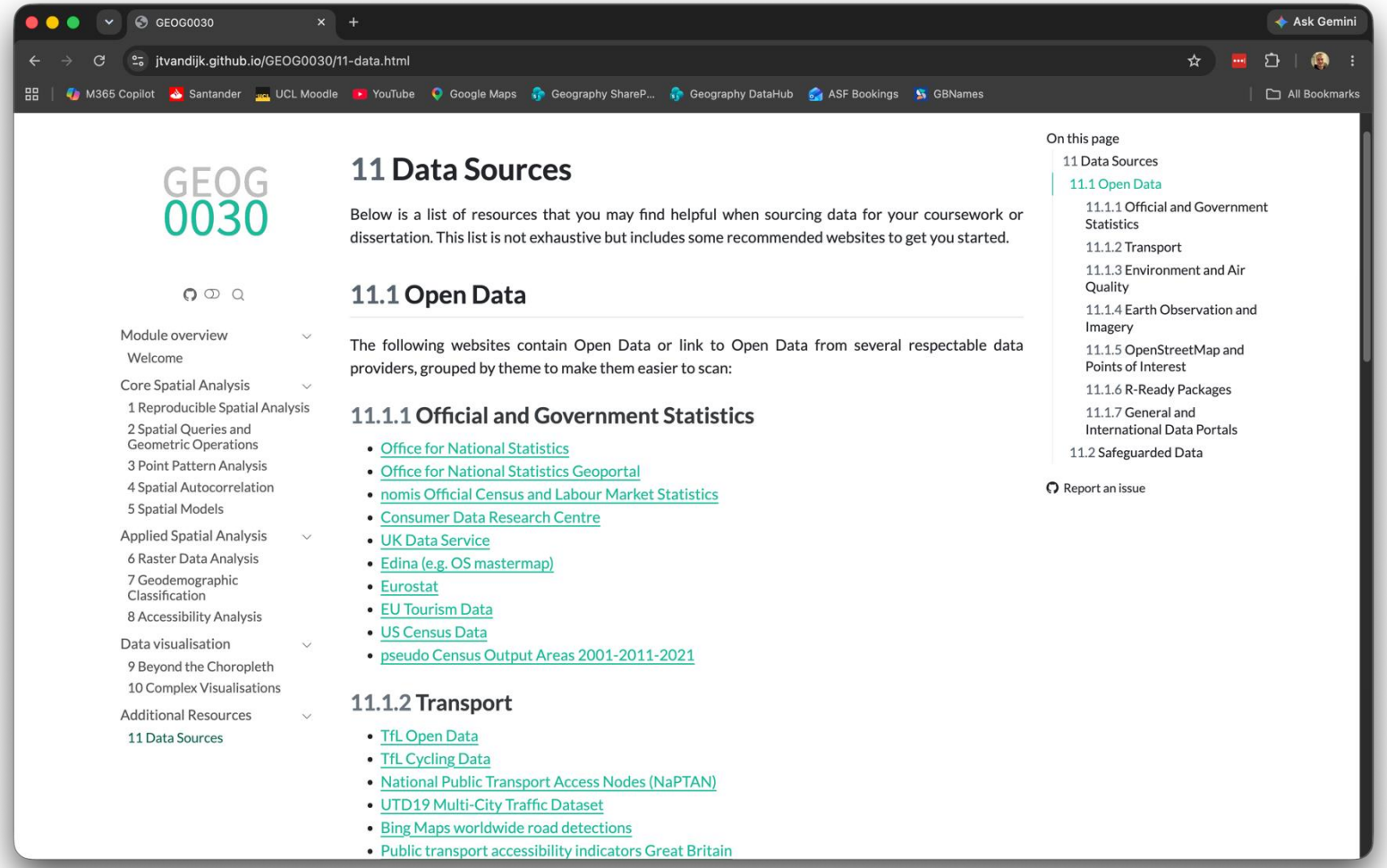
Assessment

Question	Technique	Week
What lies near something?	Buffers and spatial queries	2
Where do points cluster?	Kernel density estimation, DBSCAN	3
Are similar values clustered?	Moran's I and local Moran's I	4
Does a relationship vary by place?	Geographically weighted regression	5
What lies between my measurements?	Raster analysis and interpolation	6
Which areas are alike?	Geodemographic classification	7
How well can people reach services?	Network and accessibility analysis	8
How do I show this clearly?	Classification and bivariate maps	9 and 10

Assessment


Topic: Assessment | GEOG00
Ask Gemini
moodle.ucl.ac.uk/course/view.php?id=49768&section=4#tabs-tree-start
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
Marking criteria
For the coursework assignment, the marking criteria that will be used can be found in the Geography Coursework Marking Matrix: [Link]
For the exam, the marking criteria that will be used can be found in the Geography Exam Marking Matrix: [Link]
Instructions and examples
GEOG0030 Assessment: Social Atlas PDF
Last updated: 10-12-2025
94.8 KB
Social Atlas Example 1 - Low 2:1 PDF
1.3 MB
Social Atlas Example 2 - High 2:1 PDF
2.2 MB
Social Atlas Example 3 - Low First PDF
1.0 MB
Social Atlas Example 4 - High First PDF
5.3 MB
Assignment submission
Back to top

Assessment


GEOG0030
Ask Gemini
jtvandijk.github.io/GEOG0030/11-data.html
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
GEOG 0030
Module overview
Welcome
Core Spatial Analysis
1 Reproducible Spatial Analysis
2 Spatial Queries and Geometric Operations
3 Point Pattern Analysis
4 Spatial Autocorrelation
5 Spatial Models
Applied Spatial Analysis
6 Raster Data Analysis
7 Geodemographic Classification
8 Accessibility Analysis
Data visualisation
9 Beyond the Choropleth
10 Complex Visualisations
Additional Resources
11 Data Sources
11 Data Sources
Below is a list of resources that you may find helpful when sourcing data for your coursework or dissertation. This list is not exhaustive but includes some recommended websites to get you started.
11.1 Open Data
The following websites contain Open Data or link to Open Data from several respectable data providers, grouped by theme to make them easier to scan:
11.1.1 Official and Government Statistics
Office for National Statistics
Office for National Statistics Geoportal
nomis Official Census and Labour Market Statistics
Consumer Data Research Centre
UK Data Service
Edina (e.g. OS mastermap)
Eurostat
EU Tourism Data
US Census Data
pseudo Census Output Areas 2001-2011-2021
11.1.2 Transport
TfL Open Data
TfL Cycling Data
National Public Transport Access Nodes (NaPTAN)
UTD19 Multi-City Traffic Dataset
Bing Maps worldwide road detections
Public transport accessibility indicators Great Britain
On this page
11 Data Sources
11.1 Open Data
11.1.1 Official and Government Statistics
11.1.2 Transport
11.1.3 Environment and Air Quality
11.1.4 Earth Observation and Imagery
11.1.5 OpenStreetMap and Points of Interest
11.1.6 R-Ready Packages
11.1.7 General and International Data Portals
11.2 Safeguarded Data
Report an issue

Assessment

TL;DR story of max 1,500 words tied together by 4–6 related maps and 2–4 graphs.

Questions

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Have a good reading week!

